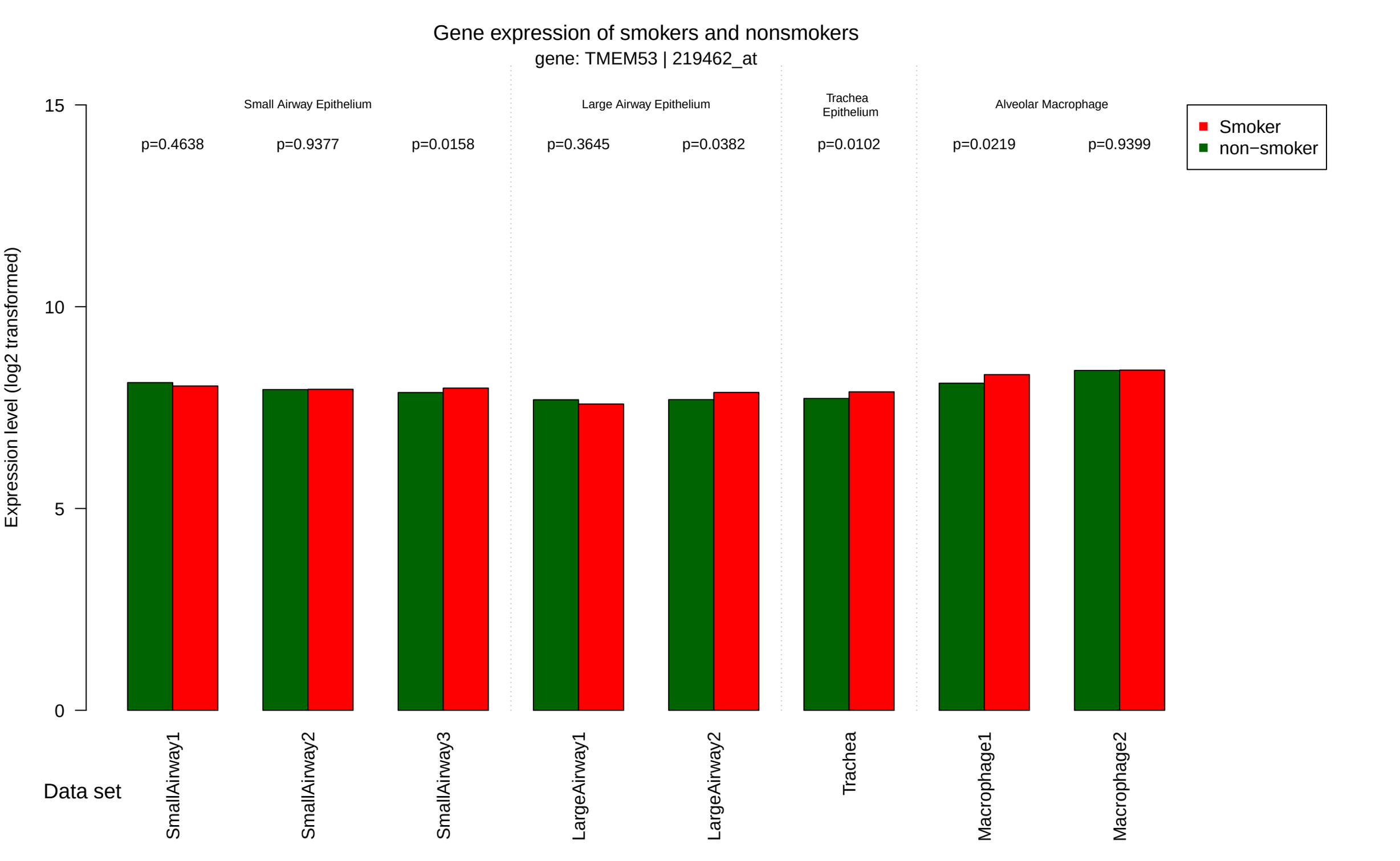
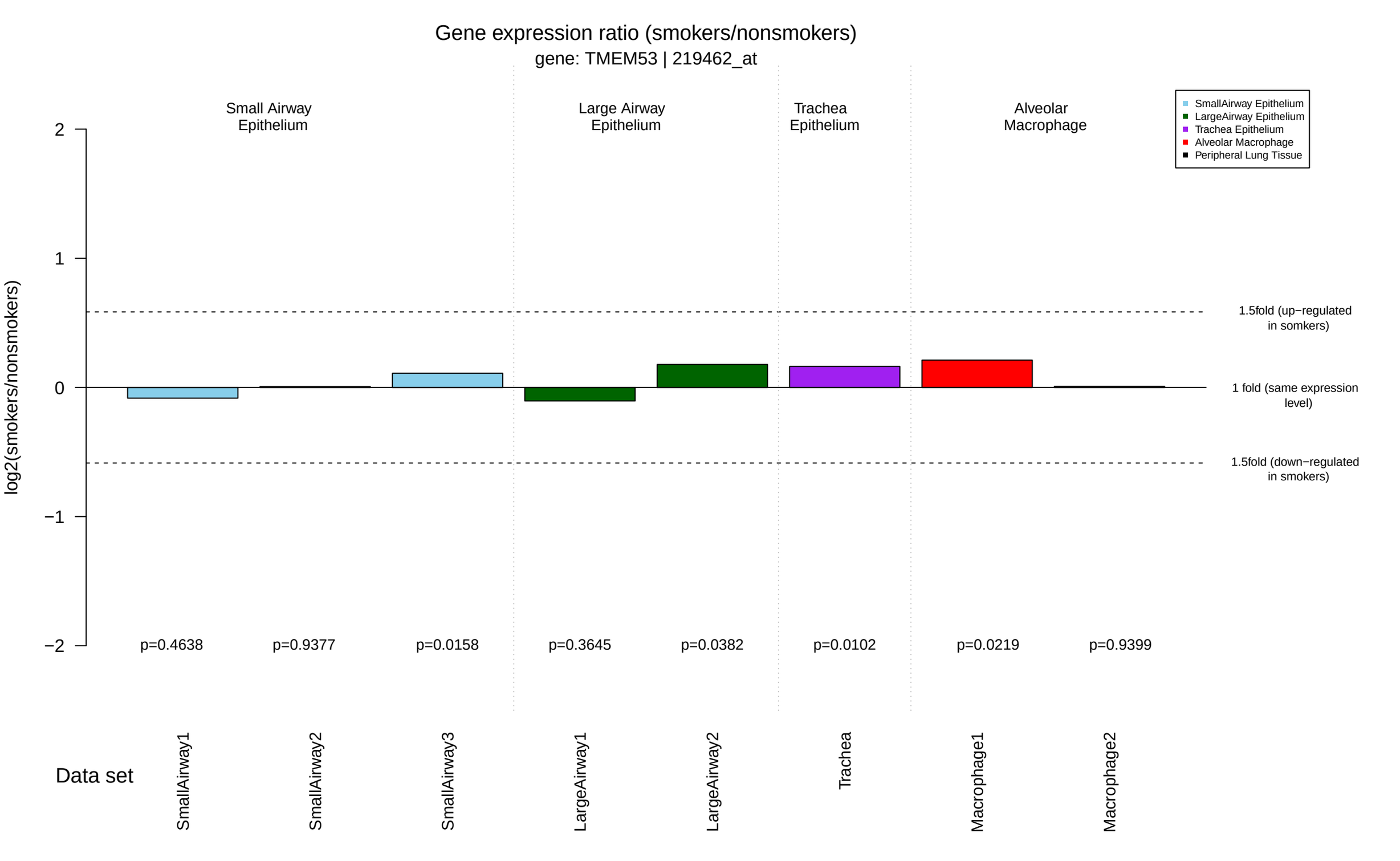
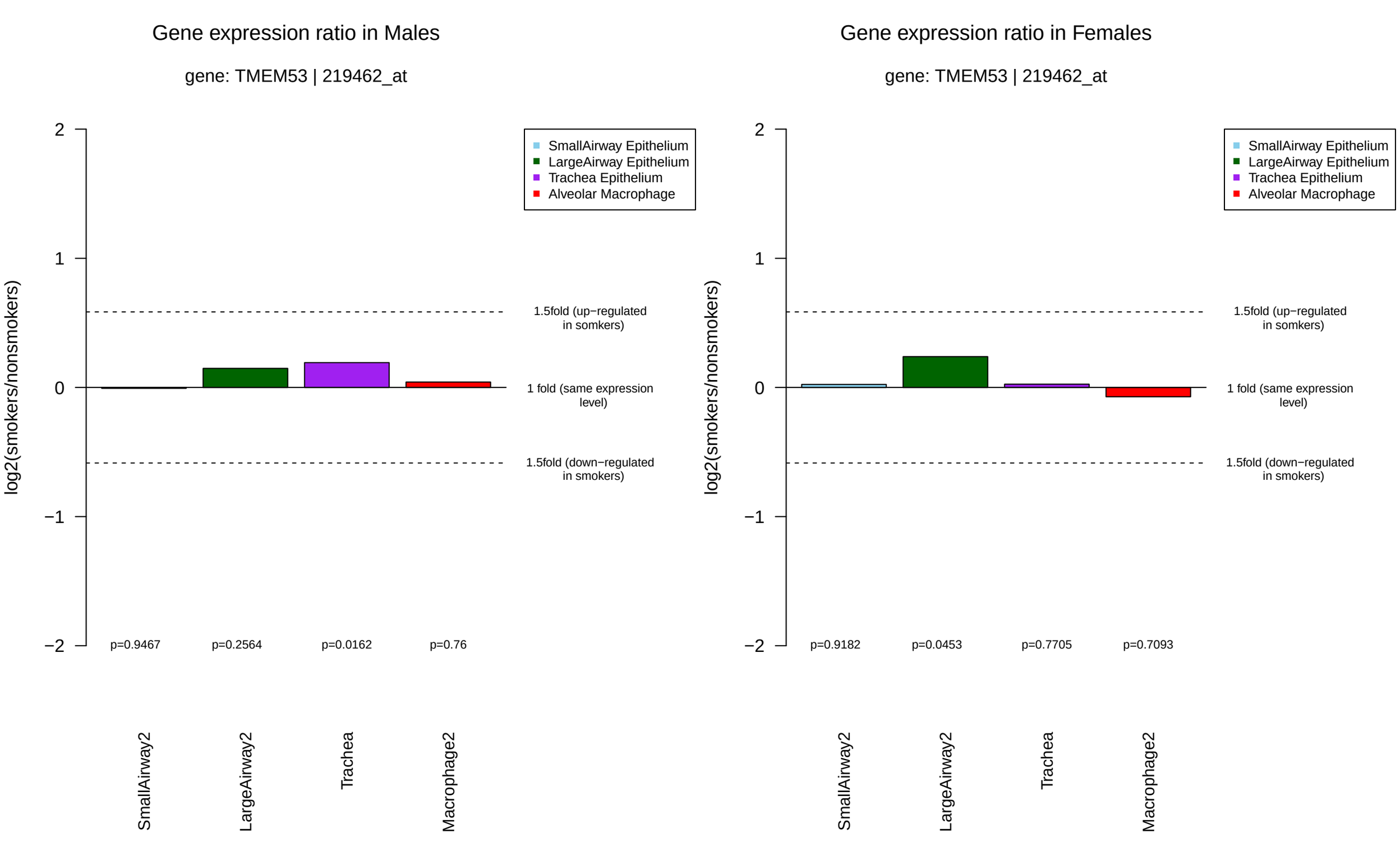
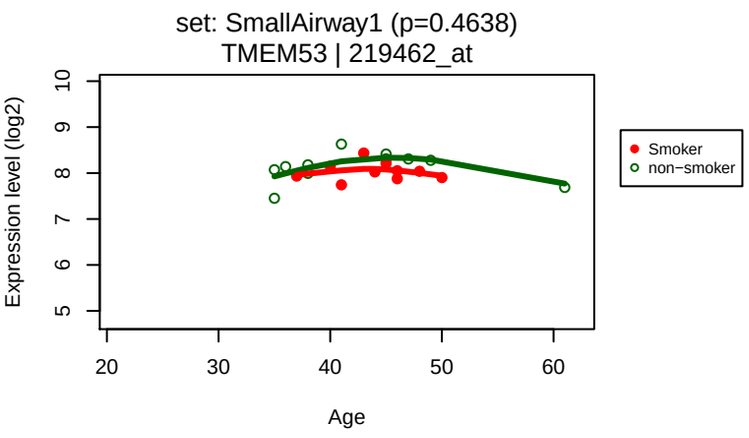
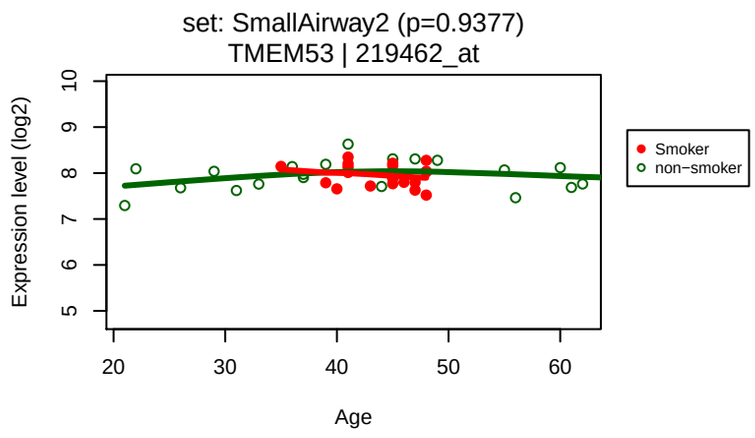
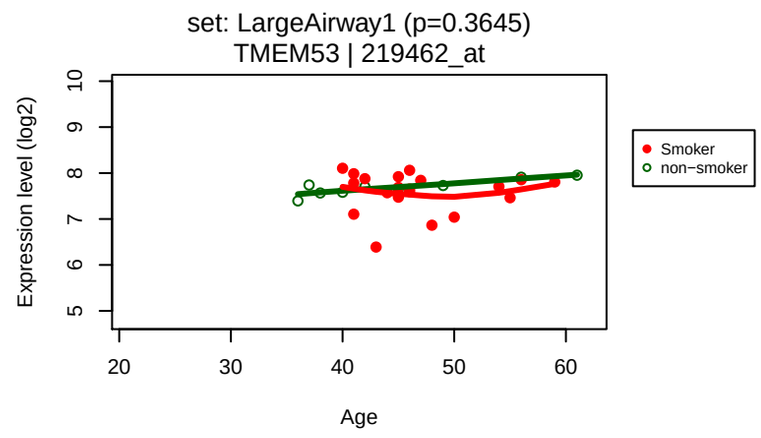

Gene expression of smokers and nonsmokers
gene: TMEM53 | 219462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4638
p=0.9377
p=0.0158
p=0.3645
p=0.0382
p=0.0102
p=0.0219
p=0.9399
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

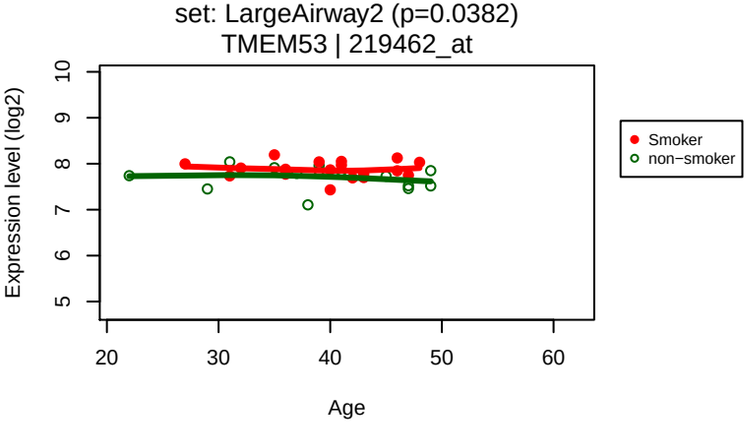

Gene expression ratio (smokers/nonsmokers)
gene: TMEM53 | 219462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4638
p=0.9377
p=0.0158
p=0.3645
p=0.0382
p=0.0102
p=0.0219
p=0.9399
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

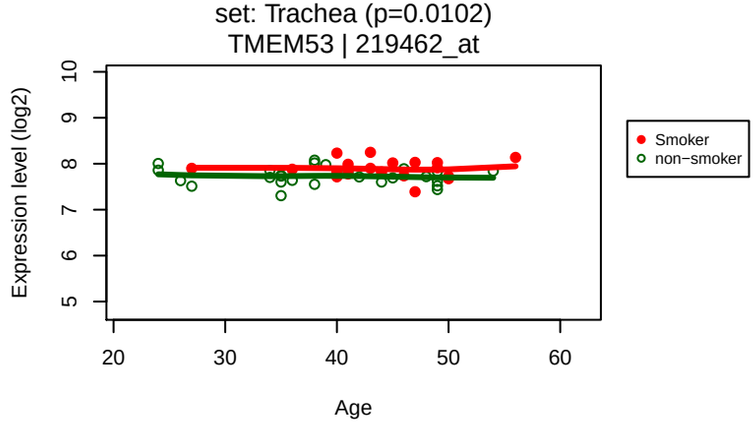

Gene expression ratio in Males
gene: TMEM53 | 219462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9467
p=0.2564
p=0.0162
p=0.76
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM53 | 219462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9182
p=0.0453
p=0.7705
p=0.7093
SmallAirway2
LargeAirway2
Trachea
Macrophage2

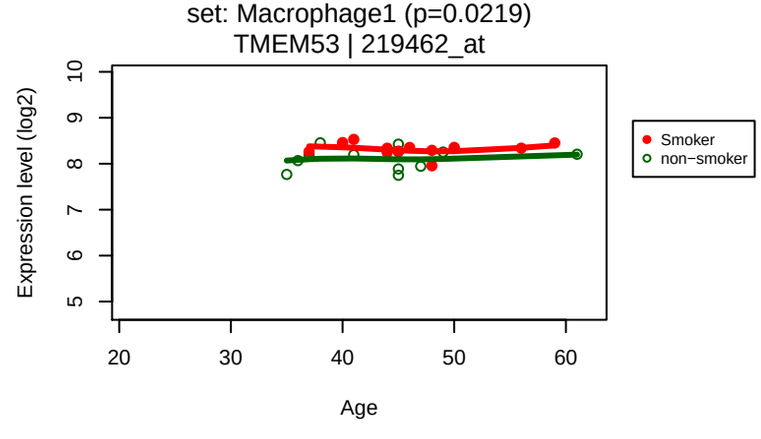

set: SmallAirway1 (p=0.4638)
TMEM53 | 219462_at
Expression level (log2)
Age
Smoker
non−smoker

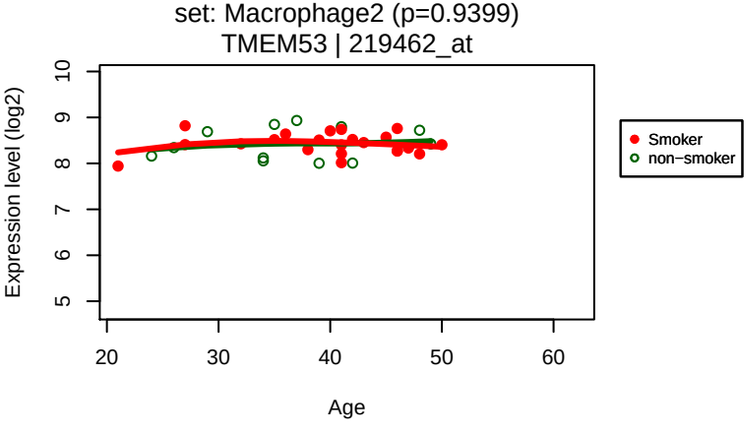

set: SmallAirway2 (p=0.9377)
TMEM53 | 219462_at
Expression level (log2)
Age
Smoker
non−smoker

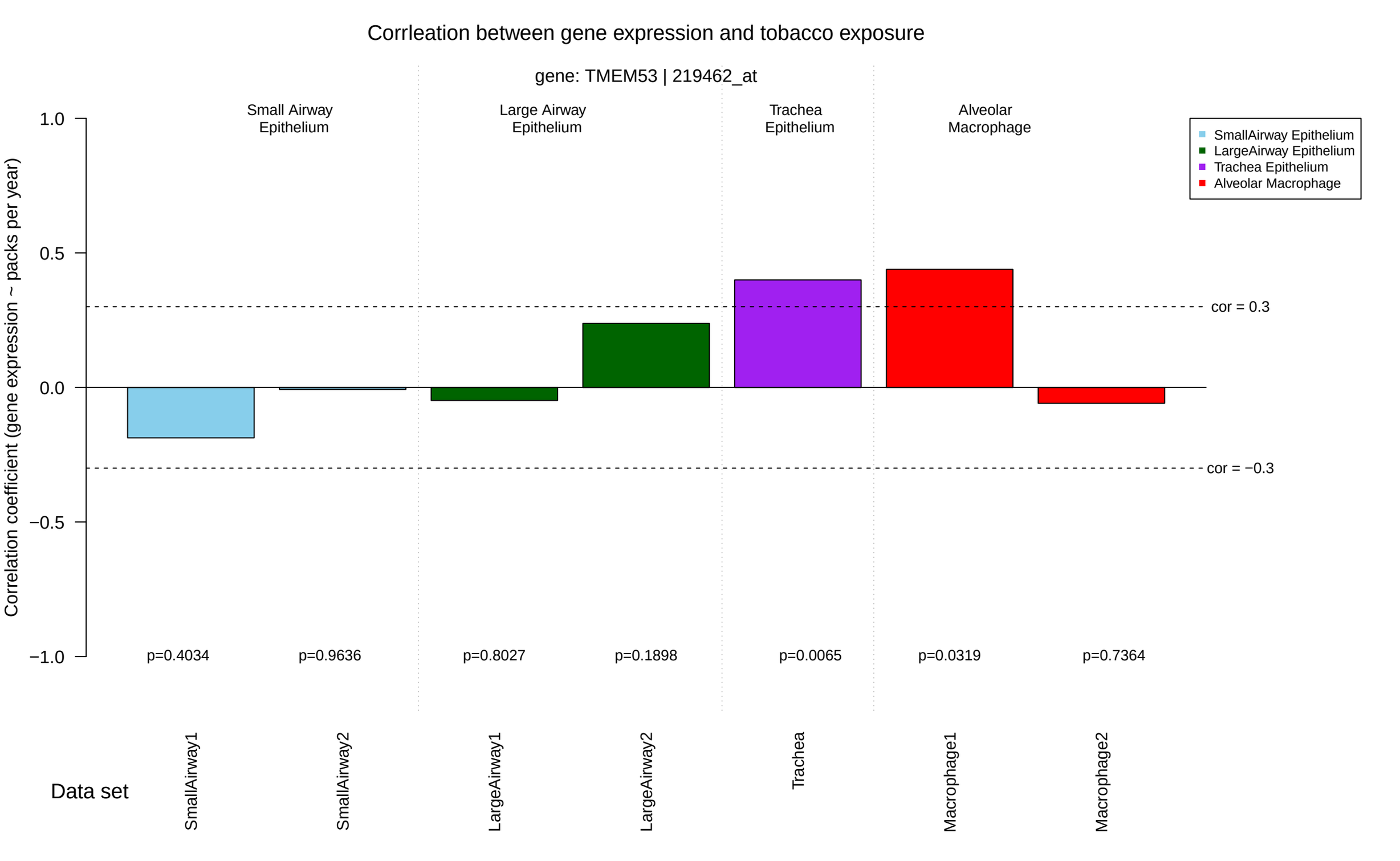

set: LargeAirway1 (p=0.3645)
TMEM53 | 219462_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0382)
TMEM53 | 219462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0102)
TMEM53 | 219462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0219)
TMEM53 | 219462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9399)
TMEM53 | 219462_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM53 | 219462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4034
p=0.9636
p=0.8027
p=0.1898
p=0.0065
p=0.0319
p=0.7364
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2